Experiment no:6

Determination and plotting of Capacitance of parallel plate capacitor in three different Dielectric medium with specific area, dielectric thickness and various dielectric constant.

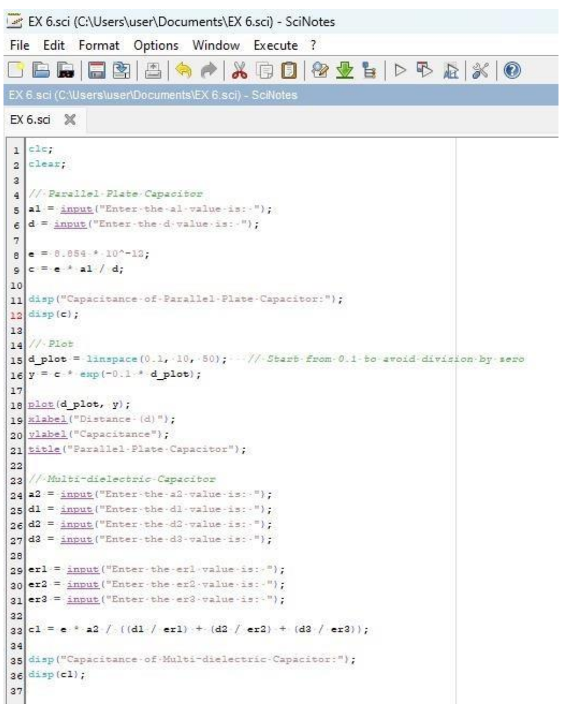
Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.

Software required:

•SCILAB version 6.0.1

Program:



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EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
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EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
EX 6.sci

clc;
clear;

// Parallel Plate Capacitor
a1 = input("Enter the a1 value is: ");
d = input("Enter the d value is: ");

e = 8.854 * 10^-12;
c = e * a1 / d;

disp("Capacitance of Parallel Plate Capacitor:");
disp(c);

// Plot
d_plot = linspace(0.1, 10, 50);   // Start from 0.1 to avoid division by zero
y = c * exp(-0.1 * d_plot);

plot(d_plot, y);
xlabel("Distance (d)");
ylabel("Capacitance");
title("Parallel Plate Capacitor");

// Multi-dielectric Capacitor
a2 = input("Enter the a2 value is: ");
d1 = input("Enter the d1 value is: ");
d2 = input("Enter the d2 value is: ");
d3 = input("Enter the d3 value is: ");

er1 = input("Enter the er1 value is: ");
er2 = input("Enter the er2 value is: ");
er3 = input("Enter the er3 value is: ");

c1 = e * a2 / ((d1 / er1) + (d2 / er2) + (d3 / er3));

disp("Capacitance of Multi-dielectric Capacitor:");
disp(c1);
```

Output:

```
Enter the a1 value is: 0.0003

Enter the d value is: 0.001

  "Capacitance of Parallel Plate Capacitor:"
   2.656D-12
Enter the a2 value is: 0.0006

Enter the d1 value is: 0.002

Enter the d2 value is: 0.003

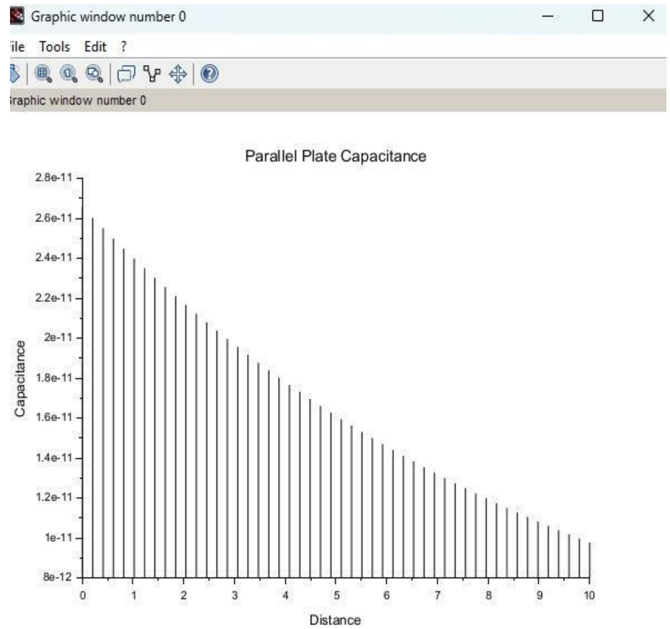
Enter the d3 value is: 0.004

Enter the er1 value is: 4

Enter the er2 value is: 6

Enter the er3 value is: 8

  "Capacitance of Multi-dielectric Capacitor:"
   3.542D-12
```



Theoretical calculation :

6.

1. Calculation for single Dielectric capacitor (C);

$$C_1 = \frac{\varepsilon_0 \cdot A}{d}$$

$$C = \frac{8.854 \times 10^{-12} \times 3 \times 1}{1 \times 10^{-3}} = \frac{2.6562 \times 10^{-15}}{1 \times 10^{-3}} \Rightarrow C = 2.6562 \times 10^{-12}$$

$$= 2.656\ \text{pF}$$

2. Calculation for Three-Layer Dielectric capacitor (C_1);

$$Dm = \left(\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} + \frac{d_3}{\varepsilon_{r3}}\right) \Rightarrow \left[\frac{4 \times 10^{-3}}{4} * \frac{3 \times 10^{-3}}{5} + \frac{2 \times 10^{-3}}{6}\right]$$

$$\Rightarrow (1.0 + 0.6 + 0.3333) \times 10^{-3}\ \text{m}$$

$$= 1.9333 \times 10^{-3}\ \text{m}$$

$$Nm \Rightarrow \varepsilon_0 \cdot A_2 = 8.854 \times 10^{-12} \times 4 \times 10^{-4} = 3.5416 \times 10^{-14}\ \text{F.m}$$

$$TC \Rightarrow (C_1) = \frac{3.5416 \times 10^{-14}}{1.9333 \times 10^{-3}}$$

$$C_1 = 1.8319 \times 10^{-11}\ \text{F} \quad \boxed{= 1.832\ \text{pF}}$$

Case 1. AIR:

```
//-Plot
d_plot = linspace(0.1, 10, 50); //-Start-from-0.1-to-avoid-division-by-zero
y = c * exp(-0.1 * d_plot);

plot(d_plot, y);
xlabel("Distance (d)");
ylabel("Capacitance");
title("Parallel Plate Capacitor");

//-Multi-dielectric-Capacitor
a2 = input("Enter the a2 value is: ");
d1 = input("Enter the d1 value is: ");
d2 = input("Enter the d2 value is: ");
d3 = input("Enter the d3 value is: ");

er1 = input("Enter the er1 value is: ");
er2 = input("Enter the er2 value is: ");
er3 = input("Enter the er3 value is: ");

c1 = e * a2 / ((d1 / er1) + (d2 / er2) + (d3 / er3));

disp("Capacitance of Multi-dielectric Capacitor:");
disp(c1);
```

Output:

```
Enter the a1 value is: 0.0003

Enter the d value is: 0.001

  "Capacitance of Parallel Plate Capacitor:"
   2.656D-12
Enter the a2 value is: 0.0006

Enter the d1 value is: 0.002

Enter the d2 value is: 0.003

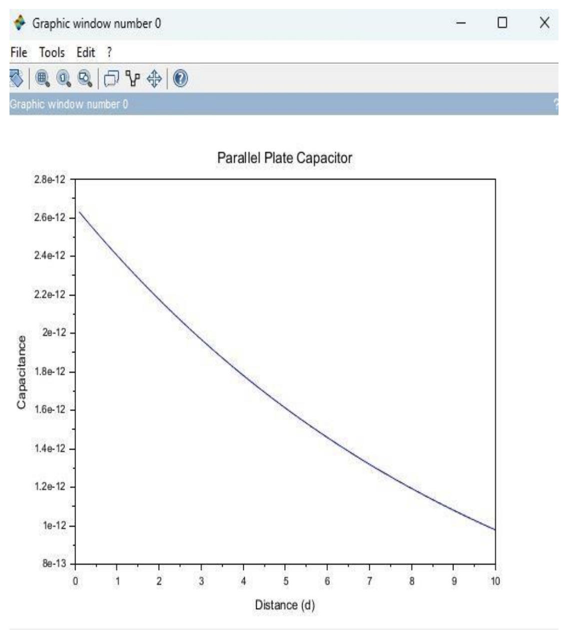
Enter the d3 value is: 0.004

Enter the er1 value is: 4

Enter the er2 value is: 6

Enter the er3 value is: 8

  "Capacitance of Multi-dielectric Capacitor:"
   3.542D-12
```



Case 2. Glass:

EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes

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EX 6.sci

```
clc;
clear;

// Constants
epsilon_0 = 8.854e-12;
area = 1e-4;        // m^2
thickness = 1e-3;   // m

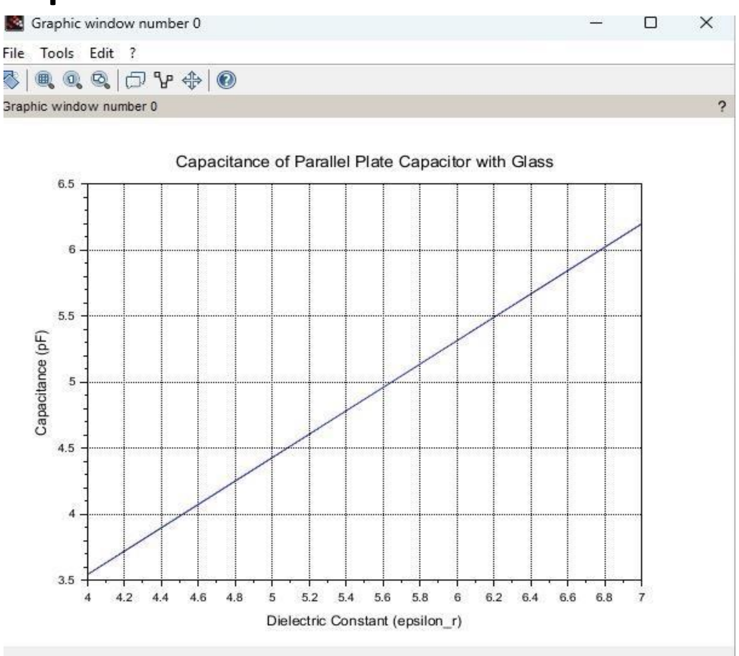
// Dielectric constant of Teflon
epsilon_r_teflon = linspace(2.0, 2.5, 100);

// Capacitance calculation
capacitance_teflon = (epsilon_r_teflon .* epsilon_0 .* area) ./ thickness;

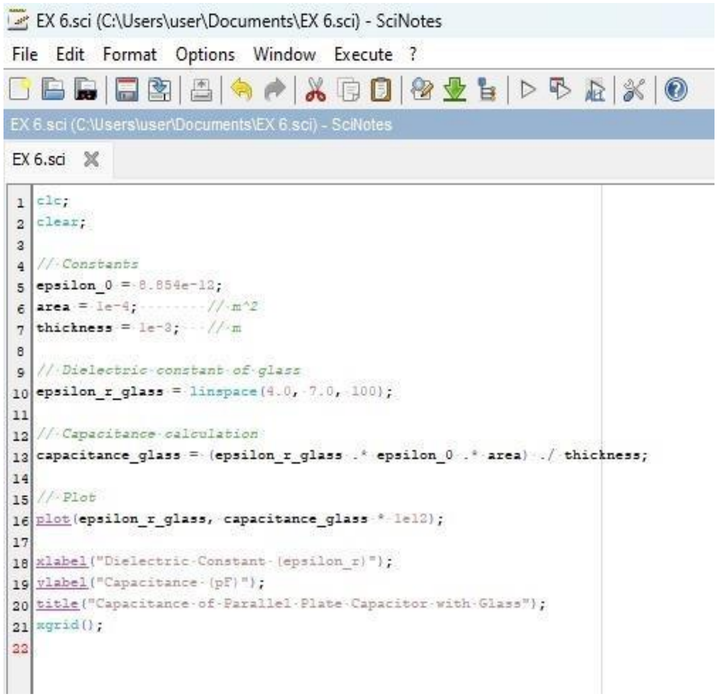
// Plot graph
plot(epsilon_r_teflon, capacitance_teflon * 1e12);

xlabel("Dielectric Constant (epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Teflon");
xgrid();
```

Output:



Case 3. Teflon:




```
clc;
clear;

// Constants
epsilon_0 = 8.854e-12;
area = 1e-4;         // m^2
thickness = 1e-3;    // m

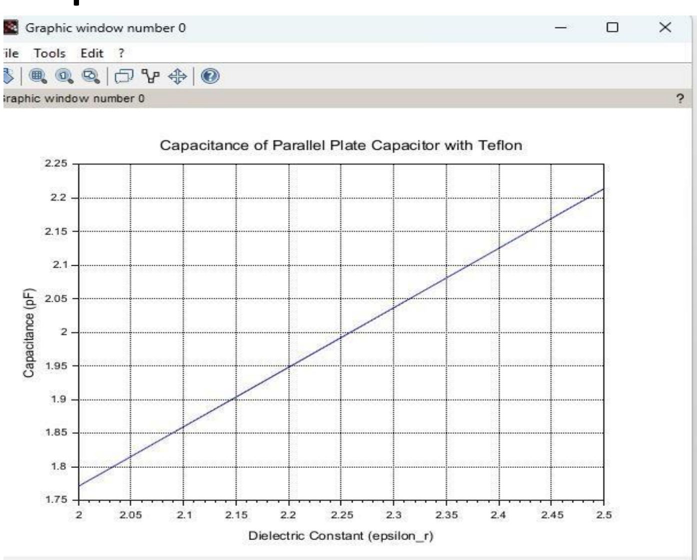
// Dielectric constant of glass
epsilon_r_glass = linspace(4.0, 7.0, 100);

// Capacitance calculation
capacitance_glass = (epsilon_r_glass .* epsilon_0 .* area) ./ thickness;

// Plot
plot(epsilon_r_glass, capacitance_glass * 1e12);

xlabel("Dielectric Constant (epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Glass");
xgrid();
```

Output:



Result :

Thus the program was verified for capacitance of parallel plate capacitor in three different dielectric media using SCILAB was successfully.